

Latent Brittleness

Measurement Dependence and Failure Heterogeneity in Neural Network Robustness

Three findings. Seed-level failure heterogeneity persists under matched specification in two regimes ($p = 6.0e-4$ MNIST, $2.0e-4$ CIFAR-10), under a preregistered design with a fixed training budget, no model selection, and a three-way disjoint split. A pure coordinate change reverses an architectural robustness conclusion on an identical pruning trajectory ($\rho = +0.63$ by surviving count, -0.30 by surviving fraction). A code-level survey of eight pruning implementations finds four non-equivalent parameterisations and three meanings of the word “fraction”.

Three self-corrections. An exploratory three-factor decomposition failed preregistered replication. A 22-predictor pre-stress screen collapsed under architecture control. A magnitude-plus-tilt curve representation is withdrawn against a monotone null.

Data, code and all preregistrations:

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Abstract

Robustness is conventionally reported as a scalar: accuracy under one corruption at one intensity. We instead measure full **degradation curves** across thirteen stressors in four perturbation families, for networks matched on architecture, training recipe and baseline accuracy, in two regimes (MNIST+MLP, CIFAR-10+CNN, 72 models each).

Failure heterogeneity persists under matched specification, in both regimes. Under a preregistered design with a fixed training budget, no model selection, and a three-way disjoint train/match/stress split, a permutation test of *no persistent cross-stressor model effect within architecture* rejects at $p = 6.0e-4$ (MNIST) and $p = 2.0e-4$ (CIFAR). Networks identical in architecture and training specification converge to parameter states with measurably and consistently different failure behaviour.

Robustness conclusions depend on the perturbation measure, within a stated boundary. On a single global magnitude-pruning trajectory, the identical physical states analysed under two coordinates give $\rho(\text{width, degradation}) = +0.63$ aligned by surviving weight count and -0.30 aligned by surviving fraction. We show this residual disagreement is structural: a pruning state is two-dimensional in (count, fraction) and architecture fixes the relation between the axes, so no single alignment holds a state fixed across differently sized networks. Reading the threshold arithmetic in eight widely used pruning implementations, we find four non-equivalent parameterisations, and — in the standard PyTorch API — a choice between the two schemes we show to disagree that is expressed as a Python type.

We report three **self-corrections**: an exploratory three-factor decomposition failed preregistered replication; a 22-predictor pre-stress screen collapsed under architecture control; and a magnitude-plus-tilt curve representation is withdrawn after its second mode

proved smaller than a matched monotone null. Separately, a family organisation of stressors reported at +0.464 under a design with test-set reuse falls to +0.173 under the clean design and is absent in CIFAR — a third instance of apparent robustness structure produced by selection rather than by the models.

1. Introduction

Two networks reach the same test accuracy. Are they equally reliable?

The question is not new. D’Amour et al. (2022) established **underspecification**: a pipeline is underspecified when it can return many predictors with equivalently strong held-out performance, and such predictors *behave differently on stress tests in a way sensitive to arbitrary iid-performance-preserving choices such as the random seed, producing variation between predictors rather than uniform degradation. We take this as our starting point, not our contribution.*

What remains open is *how* that variation is organised, and whether the instrument used to measure it is stable under choices the field does not usually state. Two conventions deserve scrutiny: reporting a scalar accuracy at a single corruption intensity, and expressing internal damage as a fraction of weights removed.

Contributions.

1. A **blocked, confirmatory** quantification of seed-level failure heterogeneity in two regimes, under a design with no model selection and no test-set reuse (§4).
2. A demonstration that a **pure coordinate change** reverses an architectural robustness conclusion on an identical pruning trajectory, with a stated scope condition derived from a cross-regime failure (§5).
3. A curve-level characterisation of degradation, with a monotone null that **retires** the two-dimensional representation we initially proposed (§6).
4. **Three documented self-corrections**, and an argument that the confounds behind them are generic to exploratory robustness analysis: architecture masquerading as latent structure (§7, §8), and test-set reuse inflating apparent robustness-space structure by roughly $2.7\times$ (§9).

1.1 Epistemic status of each claim

We state this up front because it determines how each result should be weighed.

result	status
§4 seed-level heterogeneity	confirmatory — preregistered and deposited before the data existed
§5 ϕ reversal	preregistered prediction that survived ; scope condition is post hoc
§5.2 coordinate isolation (Cohort D)	exploratory
§6 curve geometry	confirmatory failure against a preregistered null
§7, §8 negative results	§7 confirmatory failure; §8 exploratory

2. Formalism

A system is $S=(f,\theta,\mathcal{E},P)$ — observed behaviour, internal parameters, operating environment, performance functional. Under stressor s at normalised intensity ε , degradation is $D(\varepsilon) = P(0) - P_s(\varepsilon)$.

ε is a *normalised coordinate*. A **perturbation measure** φ maps it to physical damage. Removing 70% of weights and retaining a per-layer cap of 3000 weights are different φ over the same damage family. The object of study is therefore

$$D_i(s, \varphi, \varepsilon),$$

and a robustness claim that does not state φ is incomplete. Two further distinctions proved necessary and recur throughout:

- **coordinate change** — relabelling positions along one fixed path, $q \rightarrow h(q)$
- **intervention-path change** — a different path through perturbation space, $\gamma_1(q) \neq \gamma_2(q)$

Definition (latent brittleness). *Variation in degradation response remaining after conditioning on normal performance and known system specification.*

Not necessarily scalar; not necessarily universal across stressors; not invariant to how stress is defined. Each qualification is forced by a result below.

The claim under test, for architecture A and training recipe T :

$$P_i(0) \approx P_j(0), \quad \text{quad } A_i=A_j, \quad \text{quad } T_i=T_j ; ; \text{ not } \rightarrow ; ; D_i=D_j.$$

3. Method

3.1 Stressors

Thirteen stressors in four families, each acting on a different component of S : input corruption (noise, blur, occlusion, contrast), geometric shift (rotate, translate, scale), internal damage (weight drop, unit ablation), resource constraint (magnitude pruning, activation clipping), plus fixed-absolute variants of weight and unit damage for §5.

Magnitudes were calibrated once on held-out models so mean degradation at $\varepsilon=1$ lands near 0.40. Without this, saturating stressors become indistinguishable at the endpoint regardless of the path taken: in an early run, random weight drop and unit ablation returned *identical* degradation at every model because both reached chance accuracy by $\varepsilon=1$.

Perturbation RNG is seeded **independently of the training seed**, so every model sees identical corrupted inputs.

3.2 Design

The primary cohort uses a design with three properties, each fixing a documented defect in our own earlier work:

- **Fixed training budget.** 25 epochs for every model, no scanning,

no early stop. Training duration is therefore not a variable.

- **No baseline band.** All 72 models per arm retained; baseline accuracy is residualised as a covariate rather than enforced by selection. Yield is 100%.
- **Three-way disjoint split.** 30,000 train / 2,000 match / 2,000 stress. The stress set is touched only during the sweep, so no checkpoint is ever selected on it.

Two arms, one protocol. **MNIST+MLP**: width {64,128,256} × depth {1,2} × dropout {0.1,0.3}. **CIFAR-10+CNN**: channels {16,32,64} × blocks {2,3} × dropout {0.1,0.3}. Twelve cells × six seeds per arm.

Resulting baselines: MNIST 0.951–0.976 (sd 0.0058); CIFAR 0.319–0.717 (sd 0.0951). The MNIST models are competent rather than deliberately undertrained — a criticism of our earlier cohorts that this design also answers.

Blocking, rather than post-hoc regression, is what permits separate estimation of

$$B_{i,s} = \mu + A_{\text{textarch}} + G_{\text{textmodel}} | \text{textarch} + T_s + (A \text{ times } T) + (G \text{ times } T) + \text{varepsilon}.$$

4. Failure heterogeneity under matched specification

The null is stated precisely, because an earlier version of this analysis stated it wrongly twice. Permuting model labels **independently per stressor within cell** preserves within-cell spread for every individual stressor and destroys only whether the *same* models are high or low across stressors. Its null is therefore

$$H_0: \text{no persistent cross-stressor model effect within architecture cell,}$$

not $\text{Var}(\text{model}|\text{architecture})=0$. A failure to reject means a family lacks a common family-level signature; it does not imply the within-cell variation is noise, since model × stressor-specific effects with differing rankings would also fail to reject.

Result. With baseline residualised and architecture blocked, 5000 permutations:

arm	p	n	cells	min cell
MNIST + MLP	6.0e−4	72	12	6
CIFAR-10 + CNN	2.0e−4	72	12	6

Both reject. This was preregistered and deposited before the data existed.

Two networks identical in architecture and training specification, trained under an identical fixed budget, and evaluated on a stress set that played no part in their selection, converge to different learned parameter states with measurably and consistently different failure behaviour — in both regimes tested.

5. Robustness conclusions depend on the perturbation measure

5.1 The convention this concerns is prevalent, and it is not stated

The confound below is active only where damage is operationalised as a *fraction* of weights. We therefore surveyed how the field actually operationalises it, by reading source code rather than papers. Full table and sources in Appendix B.

Among reference implementations of eight widely-used pruning methods, **seven of eight compute the number of weights removed as a fraction of a parameter count**. The sample is small and was selected for influence and code availability rather than at random, so we report it as prevalence within that sample rather than as a field-wide estimate. The field additionally uses at least **four non-equivalent parameterisations** — fractional per-layer, fractional global, absolute count, and distributional (a threshold set at a multiple of a layer’s weight standard deviation) — and these are not interchangeable. Under fractional per-layer, surviving capacity scales with layer size; under fractional global, with total network size; under a distributional threshold, with the weight distribution, so the relationship to architecture is mediated by training dynamics and cannot be read off the architecture at all.

The choice is made by a keyword argument or a default value, never as a stated methodological decision. In `torch.nn.utils.prune`, the most widely used pruning API, fractional and absolute differ only by the Python *type* of the `amount` argument — `0.2` versus `200` — and nothing in the signature, documentation or tutorial indicates that this has consequences beyond convenience. The documented tutorial usage is fractional and per-layer, which is the configuration under which we observe the reversal below.

The sharpest evidence that this is treated as an implementation detail: **the same research group implemented the same method under two different scopes four years apart without noting the change**. The Lottery Ticket release code applies a percentage to each layer independently; the same authors’ later `open_lth` framework pools every layer into one vector and thresholds once. Neither file discusses the difference. We stress this is not an error — it is evidence about how the choice is regarded.

The confound requires two conditions jointly, and the conjunction matters. It is live wherever a conclusion **comparing architectures of different sizes** is drawn **under a fractional parameterisation**. Neither alone suffices: a study reporting compression ratios for a single architecture is unaffected however damage is parameterised, and a study comparing architectures under an absolute or distributional scheme is unaffected too. Seven of eight implementations here satisfy the parameterisation condition; we did not systematically record which corresponding papers satisfy the comparison condition. Doing so is what would convert prevalence of the *parameterisation* into prevalence of the *confound*, and it is the obvious extension of this survey (§11).

5.2 The comparison must change only ϕ

An earlier draft contrasted random weight deletion (fractional) with magnitude-based retention (absolute), varying the parameterisation *and* the selection rule. The clean contrast holds the selection rule

fixed — magnitude-based in both arms — and varies only ϕ . In the MNIST arm:

damage type	fractional	absolute (per-layer cap)
weights	-0.866	+0.797
units	-0.744	+0.493

Both pairs reverse, and more strongly than the earlier confounded comparison reported (-0.807 / $+0.704$). Spearman ρ between the architectural variable (width) and degradation AUC, $n = 72$.

5.3 Isolating the coordinate

To rule out that damage *allocation* rather than the coordinate produces this, we build a **single global magnitude-pruning trajectory** per model — every weight ranked once, removed smallest-first — and sample it at absolute surviving counts K reachable by every architecture. The identical physical states are then analysed aligned by K and by K/N .

Mean over the mid-range: **+0.632 by count**, **-0.302 by fraction**, reversing in every depth $\times$ dropout stratum (Figure 1). Same weights, same removal order, same states; only the cross-model alignment differs.

Surviving capacity dominates but does not exhaust the mechanism: on common support, $\rho(K, \text{deg}) = -0.828$ while $\rho(\text{width}, \text{deg} | \log K) = +0.281$ — at matched capacity, wider networks degrade *more*.

What the residual is. This is not a leftover; it is forced by the geometry of the problem, and it explains why neither coordinate is privileged. A pruning state is described by two quantities that cannot be varied independently once the architecture is fixed: surviving count K and surviving fraction K/N , related by the network’s own size N . Conditioning on K therefore does not hold the state fixed — it holds one projection fixed while the other varies systematically with width, because at matched K a wider network has necessarily shed a *larger fraction of itself*. A width-256 network at $K=20,000$ retains 7.4% of its weights; a width-64 network at the same K retains 39%. The first is far further from its trained configuration despite having the same number of surviving parameters.

So $+0.281$ is the width-dependence of *fraction removed at matched count*, and the -0.302 observed under fractional alignment is the width-dependence of *count retained at matched fraction*. They are the same two-variable system read along two different diagonals. The consequence is stronger than “report both coordinates”: **there is no alignment that holds a pruning state fixed across architectures of different sizes**, because the state is two-dimensional and architecture fixes the relation between its axes. Any single-coordinate comparison of differently sized networks under capacity damage is choosing which of two confounded quantities to hold constant, and the choice determines the sign of the answer.

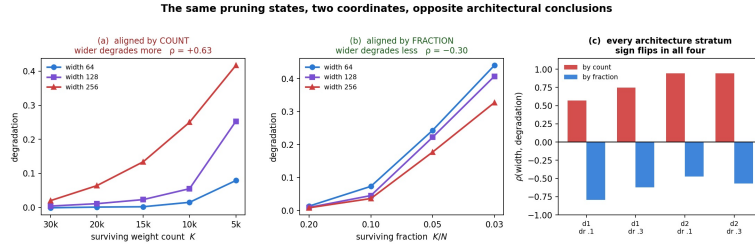


Figure 1. One global magnitude-pruning trajectory per network — every weight ranked once, removed smallest-first — analysed under two coordinates. **(a)** Aligned by surviving weight count, wider networks degrade more. **(b)** The identical physical states aligned by surviving fraction: wider networks degrade less. **(c)** The reversal holds in all four depth $\times$ dropout strata. Same weights, same removal order, same states; only the cross-model alignment differs.

5.4 Scope condition

The CIFAR arm does **not** reverse: fractional +0.382, absolute +0.542 for weights.

The mechanism is verifiable from the architectures. The absolute φ applies a per-layer cap of $k_0 = 3000(1-\varepsilon) + 150\varepsilon$, and a layer with fewer than k_0 weights is left **entirely untouched**, since $\min(k_0, \text{numel})$ retains everything. Per-layer weight counts:

cell	conv1	conv2	conv3	head	layers the cap reaches at $\varepsilon = 0.11$
ch16 / b2	432	4,608	—	320	conv2 only
ch16 / b3	432	4,608	18,432	640	conv2, conv3
ch32 / b2	864	18,432	—	640	conv2 only
ch32 / b3	864	18,432	73,728	1,280	conv2, conv3
ch64 / b2	1,728	73,728	—	1,280	conv2 only
ch64 / b3	1,728	73,728	294,912	2,560	conv2, conv3

In every CIFAR cell the first convolutional layer and the classification head lie below the cap throughout most of the ε range and are never pruned at all. Absolute damage therefore reaches only the one or two large deep convolutional layers, while fractional damage removes a proportion of *every* layer including the first. The two φ do not merely become similar — **they damage different parts of the network.**

The MLP arm is the opposite regime. Its first layer holds 50,176 to 200,704 weights and is bound hard by the cap at every ε , while its head (640-2,560) is largely untouched under both schemes. There, the two φ act on the same dominant layer and differ only in how much of it they remove.

Stated scope condition: the reversal requires that the layers dominating the parameter count lie *above* the absolute cap, so that both parameterisations act on the same weights and differ only in quantity. Where the cap exceeds a layer’s size, that layer drops out of one scheme entirely and the comparison is no longer between two measurements of one intervention. This is a limitation of the *comparison*, not evidence against measurement dependence, and it is what a two-regime test exists to expose.

We also note the reversal is demonstrated against *one* absolute ϕ — a uniform per-layer cap. Non-uniform, layer-aware budgets are a third natural ϕ we have not tested, and would not suffer this saturation.

6. Curve geometry: a claim withdrawn

Functional PCA on the degradation curves gives a magnitude mode and an early-versus-late tilt mode. We initially proposed (M,T) as an empirical coordinate system for failure.

We built a monotone null into the confirmatory design, and the claim does not survive it. Our stressors are monotone by construction with a shared origin, and fPCA on any such family concentrates variance in a ramp plus a tilt. The null is generated as follows: for each of 20 replicate datasets of the same size as the observed one, each curve takes nine independent increments drawn uniform on [0.05, 1.05], cumulatively summed from an origin at $\varepsilon = 0$ and rescaled to terminate at 1, on the same ten-point ε grid as the data. It therefore matches the empirical curves in grid resolution, monotonicity, and shared origin, and normalises endpoints rather than matching per-stressor endpoint magnitudes — a simplification we note, since the variance decomposition is computed on mean-centred curves and is scale-invariant within each replicate. Against this null:

arm	M	T	null T	T / null
MNIST	86.1%	9.1%	16.1%	0.57
CIFAR	81.4%	14.3%	16.1%	0.89

The tilt mode is **below** the monotone null in both arms. The claim that degradation curves carry meaningful early-versus-late structure is withdrawn.

The direction of the failure is itself informative and we did not anticipate it. Real degradation curves do not merely lack extra early-versus-late structure; they carry *less* of it than arbitrary monotone curves matched on endpoints. Degradation curves are therefore more **stereotyped** than chance — they cluster nearer a common shape than random monotone functions do. Combined with §4, this yields a sharper statement of what seed-level heterogeneity consists of: models sharing an architecture differ persistently and measurably in **how much** they degrade, but not detectably in **when**. The timing profile is close to a property of the stressor; the magnitude is a property of the individual trained model. This also narrows what a pre-stress predictor could hope to recover (§8): magnitude is the target with variance in it.

A defect in our own criterion, disclosed. The preregistration stated in prose that Q4 would be judged against the null. The criterion as *coded* tested $(M+T) \geq 0.90$ — the tautological part — and passed the CIFAR arm on that basis. We report Q4 as failing in both arms, following the prose rather than the code, and disclose the mismatch rather than take the favourable reading.

What survives is narrower: magnitude accounts for 81–86% of curve variance, and the equal-weight four-feature score we earlier proposed is over-parameterised. Separately, the handcrafted curvature descriptor is the most reparameterisation-sensitive of four candidate

features in both arms (rank stability 0.804 and 0.939 under $\varepsilon \rightarrow \varepsilon^2$, lowest of four), and correlates $\rho = -0.81$ with magnitude — largely redundant rather than a distinct coordinate.

7. Negative result: exploratory factor structure did not replicate

An exploratory cohort's stressor correlation matrix showed clear block structure. Parallel analysis retained three components (3.36, 2.88, 1.61 against random 95th percentiles 1.98, 1.67, 1.47); bootstrap Tucker congruence with component matching was 0.88–0.91.

Applying the **frozen** loadings to a blocked validation cohort, both preregistered kill criteria failed: loading coherence gave a PC1 gap of -0.045 $[-0.245, +0.380]$, and congruence was $0.828 / 0.681 / 0.684$ against a threshold of 0.85. **The factor result is withdrawn.**

Post-mortem: the exploratory PC1 correlates with depth (+0.60) and dropout (-0.37). The blocked cohort spans half that range on both, and the axis dissolved. **PC1 was substantially architecture.**

Two further exploratory claims died in the same audit. A rotate $\leftrightarrow$ occlusion anti-correlation (raw $\rho = -0.684$), which we had described as a robustness trade-off, falls to -0.115 ($p = 0.73$) after controlling architecture and baseline — an architectural stratum, not a trade-off. And a “width drives capacity robustness” signature proved to be the §5 artefact arriving by another route.

8. Negative result: the pre-stress predictor screen

We screened 22 baseline internal quantities — input Fisher top eigenvalue, input-Jacobian RMS norm, logit margins, relative weight-space sharpness, Hessian top eigenvalue and trace, activation participation ratio, layerwise effective rank, dead-unit fraction, spectral decay, class separation, and weight/gradient statistics — extracted **before any stress**, against per-family curve targets. Protocol fixed in advance: leave-3-cells-out grouped CV, ridge penalty selected inside training folds only.

Controlling baseline only, the result looks decisive: $R^2_{cv} = 0.830$ for capacity magnitude, $\rho = 0.931$. **Adding architecture to the control collapses it** to -0.274 . The strongest apparent predictor, activation effective rank ($\rho = +0.849$), is width leakage: $\rho(\text{effective rank, width}) = +0.523$, $\rho(\text{capacity degradation, width}) = -0.661$, and capacity damage is parameterised fractionally — the §5 artefact again.

Within cell, with architecture exactly constant, **one of 88 tests survives Bonferroni correction**: `weight_max`, $\rho = +0.442$, $p = 4.1e-4$ — a crude weight statistic rather than one of the theoretically motivated candidates. The next strongest, uncorrected, cluster at $|\rho| \approx 0.37$ – 0.43 (input Fisher top eigenvalue $+0.429$; input-Jacobian RMS $+0.429$; weight entropy -0.414 ; activation effective rank -0.394 ; Hessian top eigenvalue $+0.383$). We report point estimates without confidence intervals because the correct interval here is a cluster bootstrap over architecture cells, which we did not run for the screen; readers should treat these as ranked magnitudes rather than

estimates. The pattern — everything clustered near 0.4, nothing separating — is what an underpowered screen looks like, and is why we describe the result as inconclusive rather than null. The screen is inconclusive at five seeds per cell; detecting $\rho = 0.40$ at the Bonferroni threshold for 88 tests requires $n \approx 105$, plus 12 df for cell centring — about ten seeds per cell against the five used.

The Fisher/Jacobian hypothesis is directionally right but lands on a different coordinate than we expected: both track the *tilt* rather than magnitude. This is coherent with the literature — Novak et al. (2018) find *trained neural networks are more robust to input perturbations in the vicinity of the training data manifold, as measured by the norm of the input-output Jacobian, a local sensitivity measure, which should bound degradation onset rather than total magnitude. We do not claim this as a result; it emerged from the screen and requires independent confirmation.*

9. What replicated, and what did not

Replicated in both regimes: seed-level persistence (§4); curvature’s status as the most reparameterisation-sensitive feature (§6).

Regime-bounded: the ϕ reversal, with a mechanism (§5.3).

Did not replicate, and it is a methodological finding in its own right: family organisation of stressors. Under a design with checkpoint selection on the stress set, the within-minus-between family gap was +0.464. Under the clean design it is **+0.173 (MNIST) and −0.011 (CIFAR)**. Less than half the reported size in one regime, absent in the other. The preregistration committed in advance to reporting a weaker clean effect as the more trustworthy number; that contingency fires here.

This belongs alongside §7 and §8 as a third instance of the same class of problem. In §7 and §8, randomising architecture and then analysing residual structure recovered architecture and presented it as something else. Here, selecting checkpoints on the set later used to measure degradation inflated apparent structure in robustness space by roughly a factor of 2.7. The mechanism differs; the lesson is identical. Structure discovered in robustness data is disproportionately likely to be an artefact of how the models were selected or grouped, and the only reliable defences are blocking and holdout — not post-hoc control on the same cohort, which draws on the same variance that produced the apparent structure.

We note one property of the MNIST family structure that does survive scrutiny: it is not a coordinate artefact. Drawing **independent random monotone reparameterisations for each of eleven stressors**, the family gap exceeded criterion in 300/300 draws (median +0.483 against +0.485 for the identity coordinate). But this establishes that the *MNIST* structure is coordinate-invariant, not that it generalises.

10. Discussion

Ordinary training selects from an equivalence class defined by in-distribution performance: $\theta_1, \dots, \theta_n$ such that $P(\theta_k) \approx P_0$. That class is not an equivalence class under perturbation. The fracture is measurable and survives every control we could apply.

A methodological reading: robustness research is unusually exposed to a specific confound in which **architecture masquerades as latent structure**. It defeated us twice, in two independent analyses, and in both cases the exploratory version was more compelling than the corrected one. Randomising architecture, filtering on performance, and analysing the residual structure will recover architecture and present it as something else. Post-hoc regression on the same cohort does not catch this, because discovery and control draw on the same variance. Blocking does.

11. Limitations

1. **Regime.** Two regimes, both image classification, both small models. Whether these results hold at scale, or outside vision, is untested.
2. **The pruning survey is a sample, not a census.** Eight implementations chosen for prominence, not drawn randomly; Blalock et al. aggregate 81 papers. Proportions are indicative. More importantly, we did **not** systematically record whether each paper draws an architectural conclusion — the confound distorts a result only where networks of different widths or sizes are compared. A single-architecture sparsity result is unaffected regardless of parameterisation. This is the most valuable extension of the survey and it is incomplete.
3. **Predictor screen power.** §8 is inconclusive at five seeds per cell, not null.
4. **One absolute ϕ .** §5 tests a uniform per-layer cap, not the class of absolute parameterisations. Layer-aware budgets are untested.
5. **CIFAR baseline spread.** sd 0.0951 against MNIST's 0.0058. Baseline is residualised, but the CIFAR arm has less statistical headroom.
6. **Interaction versus measurement noise.** Stochastic damage is replicated twice per point, which helps but does not fully separate them.
7. **Q4 criterion mismatch**, disclosed in §6.

12. Related work

Underspecification. D'Amour et al. (JMLR 23(226), 2022) establish the phenomenon we build on: predictors with equivalently strong held-out performance *behave differently on stress tests in a way sensitive to arbitrary iid-performance-preserving choices such as the random seed*. Our §4 is a blocked, confirmatory, two-regime version of a phenomenon they identified — *architecture held fixed rather than varying, selection removed entirely, and the stress set disjoint from anything used to choose a model*. What is not in their work is the *measurement-dependence result (§5) and the demonstration that two apparent structural findings were architecture in disguise (§7, §8)*.

Rashomon sets. Semenova, Rudin and Parr define the Rashomon set as the set of models that perform approximately equally well on a given dataset, with the Rashomon ratio the ratio of the volume of the set of accurate models to the volume of the hypothesis space. Latent brittleness as we define it is essentially the **diameter of that set under stress**, and the connection is generative rather than merely terminological. Semenova et al. (2024) show that noisier datasets lead to larger Rashomon ratios through the way that practitioners train models. If our framing is right, this predicts that **seed-level failure heterogeneity should be larger on noisier tasks**. Our two arms differ in difficulty, and the direction is weakly consistent — CIFAR $p = 2.0e-4$ against MNIST $p = 6.0e-4$ — but the arms differ in too many respects for this to be a test. It is a prediction the framework now makes, and the obvious next experiment: hold architecture and dataset fixed and vary label noise.

Metric non-standardisation in pruning. Blalock et al. (MLSys 2020) aggregate results across 81 pruning papers and report that the community suffers from a lack of standardised benchmarks and metrics, to a degree that makes it hard to compare pruning techniques against one another or to judge how much progress the field has made over three decades. §5 sharpens this into a specific, sign-flipping instance: count versus fraction of surviving weights, on an identical trajectory. Where they show results are incomparable, we exhibit one metric pair under which a conclusion inverts.

Capacity versus sparsity. Timml et al. (2022) find that a rapid robustness drop caused by network compression observed in the literature is due to a reduced network capacity rather than sparsity. Our §5.3 is the measurement-side counterpart, and Cohort D supplies the common-support test their framing implies: at matched surviving capacity, $\rho(\text{width, degradation} \mid \log K) = +0.281$ — capacity dominates but does not exhaust the effect. The two readings imply different remedies — report multiple φ ; report effective capacity — and we recommend both.

Parameterisation in practice. Blalock et al.’s finding that metrics are not standardised is a claim about reporting. §5.3 extends it to implementation: we read the threshold arithmetic in eight widely used pruning codebases and find at least four non-equivalent schemes, three meanings of the word “fraction”, and the same authors switching scope between two releases of one method without comment. Where they show results are hard to compare, we show that one specific and widely available choice inverts a conclusion.

A competing mechanism for width-robustness reversal. Zhu et al. (2022) prove that in the under-parameterised setting width has a negative effect on robustness while it improves robustness in the over-parameterised setting — a sign reversal in the same relationship we study, from an entirely different mechanism. The two must be distinguished. Theirs is a *capacity-regime* effect: the sign changes as networks cross between parameterisation regimes, and it is a property of the networks. Ours is a *coordinate* effect: the sign changes on a single set of trained networks, on one physical pruning trajectory, with the same weights removed in the same order, when the states are aligned for comparison in two different ways (§5.3). Our per-stratum result holds depth and dropout fixed, so it is not a regime shift within

our data. But the two could compound, and we have not tested for that: a study spanning parameterisation regimes *and* reporting under one coordinate would confound them, and would be a natural place for the interaction to be measured.

Local sensitivity and degradation onset. Novak et al. (ICLR 2018) survey thousands of models and find the input-output Jacobian norm *correlates well with generalization, establishing it as a measure of local function geometry near the data manifold. This is why §8 finds Jacobian and Fisher measures tracking degradation onset rather than magnitude: they are local quantities. This yields a prediction, which we restate here because the coordinate it was originally framed in has been withdrawn (§6): Jacobian regularisation should shift the **onset** of degradation — the stress level at which measurable damage begins — while leaving total degradation magnitude comparatively unchanged. Onset must be measured directly, for instance as the smallest ϵ at which degradation exceeds a fixed threshold, rather than via a fitted tilt coordinate. Given §6, we expect the effect to be small: onset varies little across models in our data.*

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Note on verification. Findings attributed to Blalock et al., D’Amour et al., Novak et al., Semenova et al. (both), Timpl et al., and Zhu et al. were verified against source. Entries for Frankle & Carbin, Han et al., Lee et al., Tanaka et al., and Wang et al. are cited for their *reference implementations*, which we read directly (Appendix B); volume and page details for those should be checked against the published versions before submission.

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Use of AI assistance. Experimental code, statistical analysis, and manuscript drafting were carried out with the assistance of a large language model (Claude, Anthropic), under the author’s direction. The author specified the research questions, made all design decisions, determined which claims to advance or withdraw, executed the experimental runs, and takes full responsibility for the content. All preregistrations were deposited prior to the corresponding experiments, and all analysis code is included in the deposit so that every reported quantity can be independently recomputed.

Data and code availability

All raw degradation curves for five cohorts, frozen calibration anchors, three preregistration documents, the full correction history, and the complete analysis pipeline are deposited on Zenodo under concept DOI 10.5281/zenodo.22120112 (CC BY 4.0 for text, data and logs; MIT for code). SHA-256 checksums are included and verify with `sha256sum -c`. Preregistrations were deposited **prior** to the corresponding experiments; version DOIs establish the ordering.

Appendix B — pruning parameterisation survey

Eight implementations, seven methods, source-code level. Full table, line references and limitations in `PRUNING_SURVEY.md` in the deposit. Summary:

implementation	parameterisation	scope
<code>torch.nn.utils.prune</code>	either (float vs int)	either
LTH release code	fractional	per-layer
<code>open_lth</code>	fractional	global
SNIP	fractional	global
GraSP	fractional	global
SynFlow (+SNIP, GraSP, Mag, Rand)	fractional	configurable
<code>Han_prune_by_std</code>	distributional	per-layer
<code>Han_prune_by_percentile</code>	fractional	global

Appendix A — correction history

This work went through several rounds of adversarial review. The following claims appeared in earlier drafts and are withdrawn or corrected:

claim	status
Three-factor brittleness decomposition	withdrawn — failed preregistered replication (§7)
rotate↔occlusion robustness trade-off ($\rho = -0.684$)	withdrawn — architectural stratum, partial $\rho = -0.115$
“Width drives capacity robustness”	withdrawn — artefact of fractional φ (§5)
Magnitude + tilt as failure geometry	withdrawn — tilt below monotone null (§6)
Family organisation gap +0.464	corrected to +0.173 (MNIST), absent (CIFAR) (§9)
Effective capacity fully explains the reversal	corrected — positivity violation; +0.281 residual (§5.3)
Permutation null tested $\text{Var}(\text{model} \mid \text{arch}) = 0$	corrected — null restated (§4)
Curvature as “purest structural signal”	corrected — $\rho = -0.81$ with magnitude, largely redundant
Predictor screen $R^2 = 0.83$	corrected — collapses under architecture control (§8)

A full changelog with 41 numbered corrections is in the deposit.

Pruning parameterisation survey

Pruning parameterisation survey

Purpose. §5 identifies a confound that is active only where damage is operationalised as a fraction of weights. This survey establishes what parameterisations the literature actually uses, by reading implementations rather than papers.

Method. For each entry we located the line of code computing the pruning threshold and recorded (a) whether the count removed derives from a *size* or from a *fixed target*, and (b) whether it is applied per layer or across the whole network. Every entry below was read directly from source. Implementations are cited by repository and file so a reader can verify.

n = 8 implementations. A sample, not a census, and reported as such.

Results

#	implementation	file	threshold computation	class
1	PyTorch torch.nn.utils.prune	torch/nn/utils/prune.py	amount is float or int	both
2	Frankle & Carbin, LTH release	foundations/pruning.py	round(percent * sorted_weights.size)	fractional
3	Frankle et al., open_lth	pruning/sparse_global.py	ceil(pruning_fraction * n_remaining)	fractional
4	Lee et al., SNIP	snip/model.py	int(round(num_params * (1 - target_sparsity)))	fractional
5	Wang et al., GraSP	pruner/GraSP.py	int(len(all_scores) * (1 - keep_ratio))	fractional
6	Tanaka et al., SynFlow	Pruners/pruners.py	int((1.0 - sparsity) * numel())	fractional
7	Han et al. (reimpl.), prune_by_std	net/prune.py	np.std(layer_weights) * sensitivity	distributi
8	Han et al. (reimpl.), prune_by_percentile	net/prune.py	np.percentile(abs(all_alives), q)	fractional

Repositories: pytorch/pytorch; google-research/lottery-ticket-hypothesis; facebookresearch/open_lth; namhoonlee/snip-public; alecwangcq/GraSP; ganguli-lab/Synaptic-Flow; and a widely used PyTorch reimplementation of Han et al. (2015).

Entry 6 additionally implements SNIP, GraSP, magnitude and random pruning under the same threshold arithmetic, so the fractional pattern covers five methods in that file alone.

Findings

1. Fractional dominates; scope does not. Seven of eight compute the count removed as a fraction of a size. Only entry 7 does not. But *scope* splits nearly evenly — three per-layer, four global, one

configurable at runtime. Since surviving capacity relates to architecture differently under the two scopes, this matters as much as the fractional question.

2. At least four non-equivalent parameterisations are in use.

- **fractional per-layer** — count scales with each layer’s size
- **fractional global** — count scales with total network size; layer allocation emergent
- **absolute** — count independent of size (available, but not the documented default)
- **distributional** — threshold from the weight distribution; count emerges from training

The fourth is the most consequential. Under `prune_by_std`, two layers with different weight distributions retain different fractions at the same `sensitivity` setting. The relationship between architecture and surviving capacity is then mediated by whatever shaped those distributions during training, and cannot be read off the architecture at all.

3. “Fraction” denotes different quantities — but not all differences are of one kind. GraSP’s `ratio` is a prune fraction (`keep_ratio = 1-ratio`); SynFlow’s `sparsity` is a keep fraction. These are **algebraically invertible**: one-shot, and producing identical retained sets under the appropriate substitution. They are notational variants, not distinct parameterisations — though SynFlow’s naming is inverted relative to its meaning, since `sparsity` denotes the fraction *retained*, as its own consistency check (`remaining_params - total_params*sparsity`) confirms.

`open_lth`’s `pruning_fraction` is different in kind: a fraction of *currently remaining weights*, applied iteratively. The retained set after *n* rounds depends on the schedule and not only on the terminal density, so it does **not** reduce to a one-shot fraction. Two notational variants and one functionally distinct scheme. An earlier version of this survey described these as “three meanings”, conflating the two kinds of difference.

4. The choice is a type annotation in the standard API. PyTorch’s pruning functions take `amount` as int or float: float is a fraction, int an absolute count. The two parameterisations whose disagreement §5 documents are therefore the same function call with a different Python type, and nothing in the signature indicates the choice can invert an architectural conclusion. The official tutorial demonstrates the fractional form applied per module.

5. The clearest evidence that the choice is treated as an implementation detail. The same research group implemented the same method under two different scopes. The LTH release code (entry 2) applies a percent to each layer independently; `open_lth` (entry 3), the authors’ later framework, concatenates all layers and thresholds once. Neither file mentions the difference, and neither is wrong. The choice simply is not treated as methodological.

Not “the field uses the wrong parameterisation” — none of these is wrong. The defensible claim is:

The parameterisation of internal damage is not standardised and is not treated as a methodological choice. Across eight implementations we find at least four non-equivalent schemes, a functionally distinct iterative scheme alongside two notational variants of “fraction”, and one case of the same authors switching scope between two releases of the same method without comment. In the most widely used pruning API, the choice between the two parameterisations whose disagreement we document is a Python type.

*The confound identified in §5 requires **both** conditions to be met: it is active wherever a conclusion **comparing architectures of different sizes** is drawn **under a fractional parameterisation**. Neither alone suffices. A study reporting compression ratios for a single architecture is unaffected however it parameterises damage; a study comparing architectures under an absolute or distributional scheme is unaffected too. Seven of eight implementations in this sample satisfy the second condition; we did not systematically record the first, which is the obvious extension.*

Limitations

- **n = 8.** Proportions are indicative, not estimates.
- Entries were selected for influence and code availability, not at random. Highly cited work with public code is over-represented.
- Two entries are third-party reimplementations (Han et al.; SynFlow’s versions of SNIP and GraSP), though for GraSP we verified the original separately and it matches.
- We did not systematically record whether each *paper* draws an architectural conclusion. That column determines where the confound is live rather than merely present, and is the obvious extension.